

Compact Laser MBE System

(Model: PAC-LMBE)

Maintenance Manual



Pascal Co., Ltd.

Headoffice 1-16-4 Showa-cho, Abeno-ku, Osaka, 545-0011 Japan

TEL : +81-6-6626-1321 FAX : +81-6-6626-1323

< URL > <http://www.pascal-co-ltd.co.jp>

< e-mail > pascal@pascal-co-ltd.co.jp

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1 Introduction

1.1 About this document

This document, PLD maintenance manual, describes maintenance procedure or trouble-shooting of each part in PLD system. There are three kind of documents for the PLD system. One is the system manual that describes structure and function of each part/unit in the PLD system. Another is the operation manual. The last is maintenance manual.

Some units in the PLD system have occasions to injure operators or to damage the lab facility when user mistakes. See following chapters and other documents which show functions of individual units and parts in detail.

1.2 Caution

	Class 4 laser equipments are used in the PLD system. Keep away from the laser equipments and its beam path during laser emission. Wear protection glasses and clothes if user has to observe inside of the chamber or approach the laser beam path.
	X-ray is generated when an electron beam is irradiated to a substrate material with RHEED (Reflection High Energy Electron Diffraction) gun. Keep away from viewing ports which have no protection.
	Numerous electric power lines with large current or high voltage and sensor signal cables are wired at/to terminals or connectors. Do not touch the lines/cables while the PLD system runs. Shut down the electric power when user has to do cable works.
	As for an optional use of harmful or toxic gases, our original system does not equip any safety protections including a gas detector. On the occasion of use these kinds of gases, a leakage check and accompanying measure against accidents should be done thoroughly by the user.
	This system cannot sustain a normal operation under lack cooling water flow rate. An inspection of the compressed air and cooling water is regularly required.
	This system basically requires being electrified continuously. Connecting the system to a power line of insufficient capacity may cause an accidental electric power failure or system malfunction. The rating of electric power line should be sufficient.
	We do not undertake a maintenance service or a repairing in the condition that the equipment contains harmful, toxic substances or products and specified chemical substances etc. These substances should be completely removed prior to the services.

2 Maintenance period

The PLD system has various kind of components and parts. The many components/parts causes difference in degradation by operation. Pascal prescribes maintenance period or frequency to keep system performance as follows.

2.1 Checking system status at every time of operation

(Strongly recommended)

- (1) Main chamber: background pressure (see also Chapter 3 in operation manual).
- (2) Turbo molecular pumps: rotation speed, load wattage or temperature in normal status.
- (3) Laser-diode for substrate heating: performance parameter like device temperature or lifetime or others.
- (4) Chiller for laser-diode: performance parameter like temperature and water flow-rate.

2.2 Elimination of contamination by deposition material

- (1) Ablation laser port: rotation of internal quartz plate (see Sec. 3.1).
- (2) Ablation laser port: exchange of internal quartz plate (see Sec. 3.2).
- (3) Other parts.

2.3 Mechanical factor

Maintenance procedure for mechanical items is at expert level of technical difficulty. Ask Pascal engineer when user has questions or obscurities.

- (1) Gear meshing.
- (2) Other parts like spring.

2.4 Others

- (1) Chiller for laser diode: refill cooling water with adding corrosion inhibitor.
- (2) Consumption components/parts with lifetime like RHEED e-gun filament.

2.5 Machine/component maintenance by vendors' regulation

- (1) Ablation laser (Excimer laser).
- (2) Laser-diode module
- (3) Rotary pumps.
- (4) Turbo molecular pumps.
- (5) Others.

Maintenance timeline

Maintenance Issue with corresponding Sec. #, 2.x	Maintenance Period						Consumption/ Finite lifetime
	Every Time	1 Month	3 Months	6 Months	1 Year	2 Years or over	
2.1 System check							
2.2 Ablation laser port (Rotation)	(Depend on operation condition)						
2.2 Ablation laser port (Exchange)		(Depend on operation condition)					
2.3 Gear meshing		(Seasoning)				Expert level	
2.3 Stabilizer in target manipulator					Expert level		
2.4 Chiller: water refill							
2.4 RHEED filament							
2.4 RHEED fluorescent screen							
2.4 UV laser optics/IR laser optics							
2.5 Ablation laser (Excimer) gas-fill							
2.5 Rotary pump oil refill							
2.5 Laser-diode module							
2.5 turbo molecular pump							

3 Maintenance procedure

3.1 Long-time use of ablation laser port with rotating internal quartz plate

In the PLD system the ablation laser port has internal quartz plate with magnet-coupling rotation mechanism for long-time use against deposition contamination. Rotation operation of the quartz plate is described as follows.

- (1) Stop laser emission to approach laser beam path.
- (2) Open a laser-shade with canceling interlock function in PLD control software (see Figure 1).
- (3) Move outer-handling magnets to rotate the internal quartz plate until clear part of the quartz plate appears (see Figure 2).

Position of the internal quartz plate is fixed by internal magnets.

- (4) Recover the laser beam path with reference to description item 2.

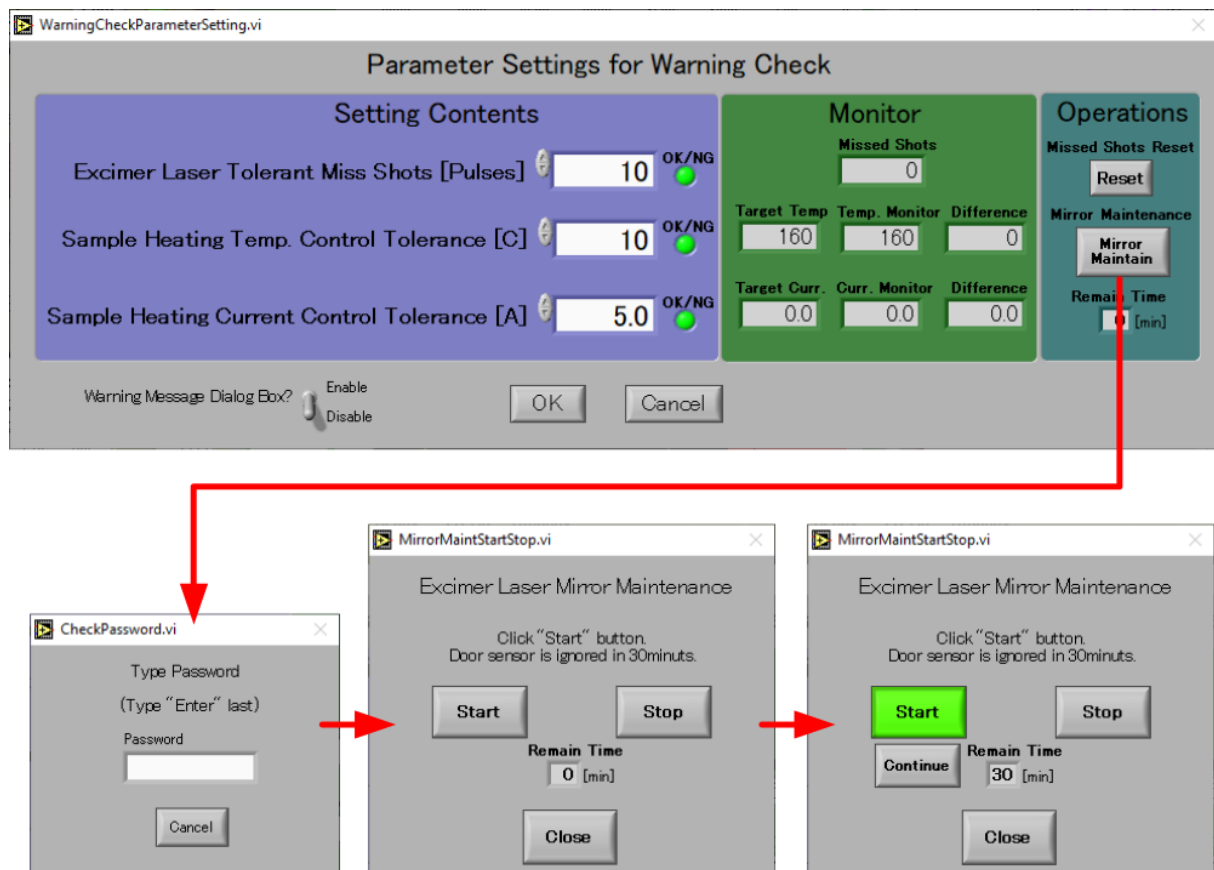


Figure 1: Operation steps to open the laser shield on PC control software.

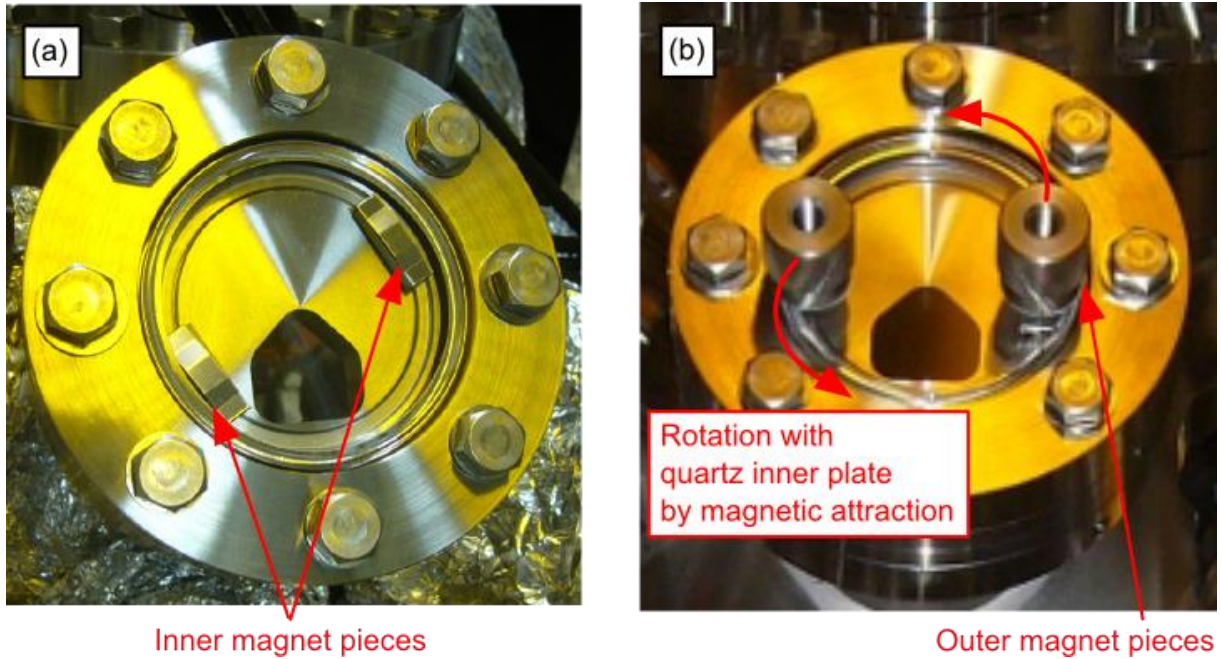


Figure 2:Rotation of internal quartz plate.

3.2 Exchange of internal quartz plate in ablation laser port

User can exchange the internal quartz plate to new one without breaking UHV in the main chamber when user wastes the clear part of the internal quartz plate in the ablation laser port as described in section 3.1.

- (1) Stop laser emission to approach laser beam path.
- (2) Open a laser-shade with canceling interlock function of the control software.
- (3) Close the valve GV2 which isolate ablation-laser port part from the main chamber.
- (4) Open the valves CV2 and CV3. Make vacuum path between the load-lock chamber and the ablation laser port.
- (5) Vent the load-lock chamber (see section 4.2 in operation manual) with above vacuum path.
- (6) Exchange the internal quartz plate with opening/closing viewport in ICF114 (CF4.5") flange.
- (7) Start vacuum pumping in the load-lock chamber (see section 4.2 in operation manual) with the ablation port vacuum path.
- (8) Open the valve GV2 after checking vacuum pressure and vacuum leak.
- (9) Close the valves CV2 and CV3.
- (10) Recover the laser beam path.

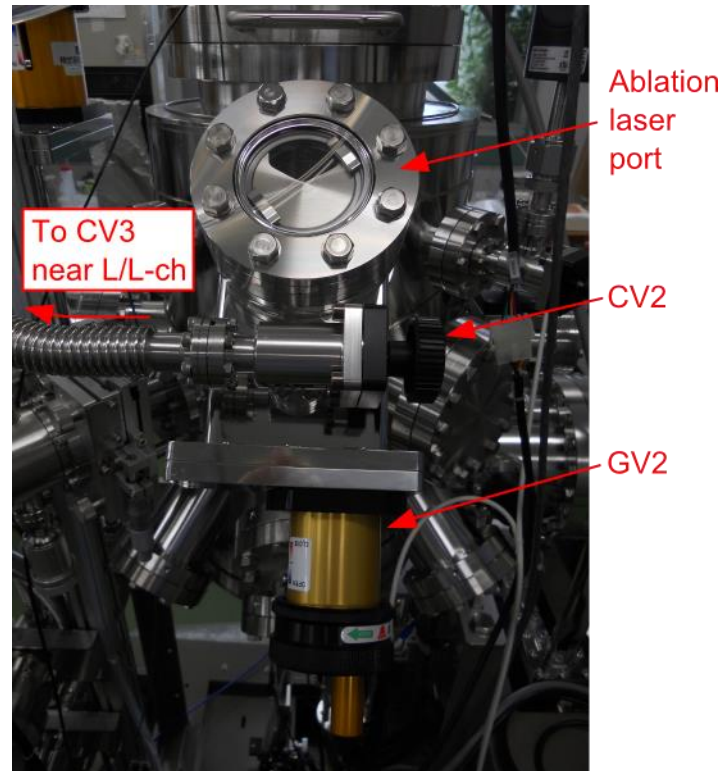


Figure 3: Vacuum pumping line near ablation laser port.

3.3 Chamber baking

- (1) Remove the laser fiber of the laser-diode substrate heating unit. Then attach an end cap of the laser fiber (see Figure 4).
- (2) Remove the lens hood of the laser-diode substrate heating unit by loosening two fixing screws to avoid thermal damage during chamber baking. And remove the sensor part of the pyrometer. Check the mount position of the pyrometer sensor part by ink-marking when removing (see Figure 4).
- (3) Remove the control/sensor cables and pipes near the chamber near the laser-diode substrate heating unit (Figure 4), the target manipulator unit (Figure 5) or the chamber to avoid thermal damage. But leave the pump (TMP) cable and the sensor cable of the vacuum gauge (IG-Main) to check pressure during chamber baking.
- (4) Check the connection of a baking heater line which is sheath heater at chamber side with white cable extension. And check the temperature monitor cable connection and temperature sensor mount at side of the chamber.

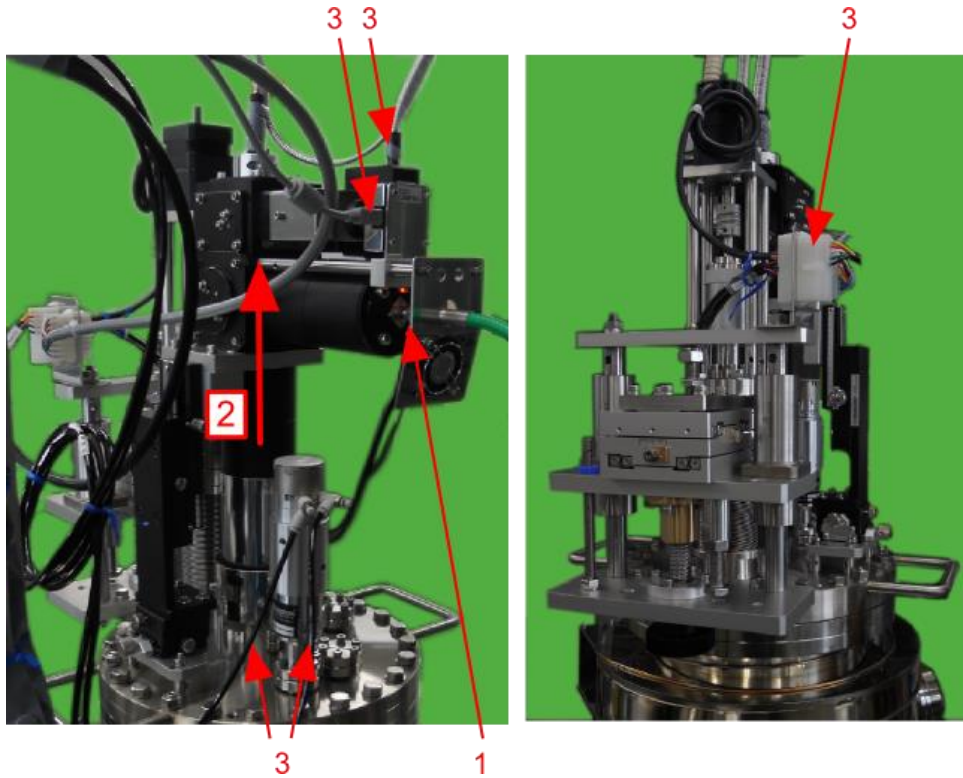


Figure 4: Removal of parts and cables/pipes near IR laser diode heating unit.

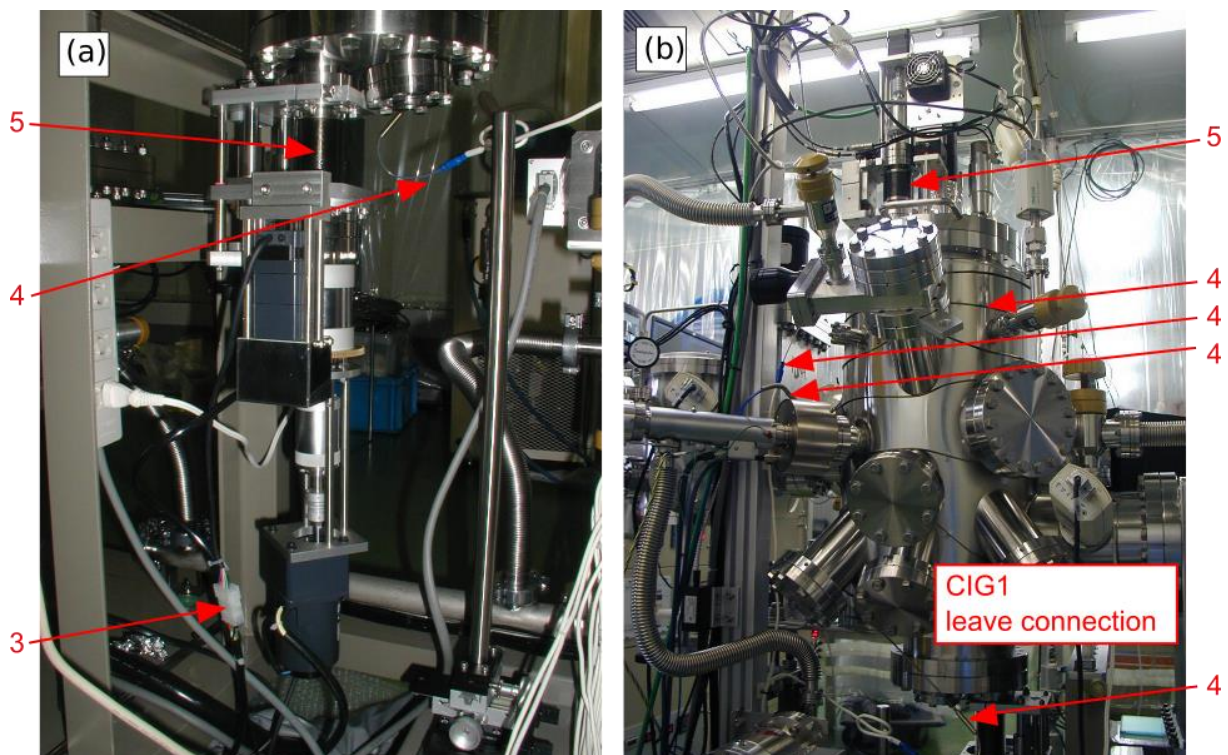


Figure 5: Preparation for chamber baking (a) near target manipulator and (b) around chamber.

- (5) Add ribbon heater line with focusing on bellows or ports or other parts as cool spots disappear during chamber baking. And add ribbon heater line if you want to bake load-lock (L/L) chamber.
- (6) Attach aluminum foil to heat the chamber uniformly.
- (7) Confirm that the power supply and the chiller for the laser-diode is turned off.
Power inlet line for the baking heater is exclusively utilized to laser-diode with respect to electrical capacity.
- (8) Adjust chamber baking temperature with baking heater controller on the panel of the control rack (see Figure 6).
- (9) Set turn-off time in the baking heater control part (see Figure 6).
- (10) Turn on the baking heater by pushing the button in the baking heater control part (see Figure 6).
- (11) The baking heater automatically turns off at set off-time. Don't move the units/parts like shutters or motorized mechanisms in the PLD chamber during the baking.
- (12) Recover the PLD system by mounting parts and connection of the cables/pipes after chamber baking. And do "homing" procedure to go back to origin (zero) position for motorized mechanisms (see item (2) of sec 3.6 in system manual).

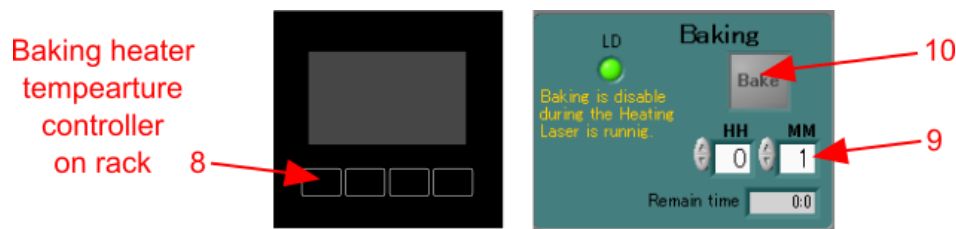


Figure 6: Baking heater control part of vacuum controller sub-window in PC software.

4 Status monitor in PLD system

System status during operation can be monitored by data logging function on PC control software as described in operation manual. Log file, which is saves in comma separated value (CSV) format, has 63 data fields to record operation status in the PLD system. The data fields #1 to #45 records current status like motorized motion position, temperature, pressure or others in numeric expression as user can easily understand. List of the data fields are shown in Table 1.

Table 1:List of data fields #1-#45 in log file.

No.	Data field	Parameter	Stored value
1	Time	Current time	HH:MM:SS
2	TG No.	Selected target number	1 to 7
3	TG Rotate	Angle position of manipulator	0.0 to 360.0 deg
4	TG Spin Speed	Target spinning speed	0.0-20.0 pps
5	TG-Z	Target height position	0.0 to 40.0 mm
6	Mask1	Mask1 position	0.0 to 160.0 mm
7	Mask2	Mask2 position	0.0 to 160.0 mm
8	Substrate	Angle position of sample stage	0.0 to 360.0 deg
9	FocusLns	Focus lens position	0.0 to 100.0 mm
10	MirrorPs	Mirror exchange position	0.0 to 100.0 mm
11	ATN	Attenuator angle position	0.0 to 45.0 deg
12	BTFvalve	Butterfly valve position	0.0 to 100.0%
13	LaserHz	Repetition frequency of ablation laser	1 to 50 Hz
14	LaserPuls	Oscillation pulse number of ablation laser	1 to 2 ³² -1 pulses
15	Laser moni	Monitored pulse number of ablation laser	1 to 2 ³² -1 pulses
16	Laser set	Set pulse number of ablation laser	1 to 2 ³² -1 pulses
17	MissedPuls	Miss-shot number of ablation laser	1 to 2 ³² -1 pulses
18	PwrMeter	Monitored power of ablation laser	0.0 to 500.0 W
19	HT set	Set electric current for sample heater	0.0 to 50.0 A
20	HT moni	Monitored current for sample heater	0.0 to 50.0 A
21	Delta HT curr	Current tolerance for sample heater	0.0 to 50.0 A

22	HT Temp set	Set temperature value for heater control	0.0 to 1400.0 degC
23	HT Temp moni	Monitored temperature value	0.0 to 1400.0 degC
24	Delta Temp	Temperature tolerance	0.0 to 1400.0 degC
25	MFC1 set	Set flow-rate for MFC1	0.00 to 100.00 sccm
26	MFC1 moni	Monitored flow-rate for MFC1	0.00 to 100.00 sccm
27	Delta MFC1	Flow-rate tolerance for MFC1	0.00 to 100.00 sccm
28	MFC2 set	Set flow-rate for MFC2	0.00 to 100.00 sccm
29	MFC2 moni	Monitored flow-rate for MFC2	0.00 to 100.00 sccm
30	Delta MFC2	Flow-rate tolerance for MFC2	0.00 to 100.00 sccm
31	MFC3 set	Set flow-rate for MFC3	0.00 to 100.00 sccm
32	MFC3 moni	Monitored flow-rate for MFC3	0.00 to 100.00 sccm
33	Delta MFC3	Flow-rate tolerance for MFC3	0.00 to 100.00 sccm
34	MFC4 set	Set flow-rate for MFC4	0.00 to 100.00 sccm
35	MFC4 moni	Monitored flow-rate for MFC4	0.00 to 100.00 sccm
36	Delta MFC4	Flow-rate tolerance for MFC4	0.00 to 100.00 sccm
37	MFC5 set	Set flow-rate for MFC5	0.00 to 100.00 sccm
38	MFC5 moni	Monitored flow-rate for MFC5	0.00 to 100.00 sccm
39	Delta MFC5	Flow-rate tolerance for MFC5	0.00 to 100.00 sccm
40	Prc Pres Main	Process-gas pressure in Main chamber	
41	Vac Pres Main	Vacuum pressure in Main chamber	
42	Back Pres Main	Vacuum pressure at TMP/Main backward	
43	Prc Pres L/L	Process-gas pressure in L/L chamber	
44	Vac Pres L/L	Vacuum pressure in L/L chamber	
45	Back Pres L/L	Vacuum pressure at TMP/LL backward	

Data fields from #46 to #63 show control status in the PLD system by 16 bits binary expression (0x0000 to 0xFFFF): on (1)/off (0) state for each control factor. User should extract a bit status by multiplying a mask bit to read out the on/off state. List of the data fields are shown in Table 2.

Table 2:List of data fields #46-#63 in log file.

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
46	Heat Stat	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	(Reserved)
		12	(Reserved)	4	(Reserved)
		11	(Reserved)	3	ON/OFF monitor in PS
		10	(Reserved)	2	Ramp rate setting
		9	(Reserved)	1	PID Heater control
		8	(Reserved)	0	Heater power supply (PS)

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
47	Depo Laser Stat	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	(Reserved)
		12	(Reserved)	4	(Reserved)
		11	(Reserved)	3	(Reserved)
		10	(Reserved)	2	Gate shutter (Reserved)
		9	(Reserved)	1	HV in excimer laser (Reserved)
		8	(Reserved)	0	Inter-locking in excimer laser

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
48	Pump Stat	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	TMP4 (2nd RHEED)
		13	(Reserved)	5	TMP3 (L/L)
		12	(Reserved)	4	TMP2 (RHEED)
		11	(Reserved)	3	TMP1 (Main)

		10	(Reserved)	2	DP3 (L/L)
		9	(Reserved)	1	DP2 (2nd RHEED)
		8	(Reserved)	0	DP1 (Main)

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
49	Valve Stat1	15	(Reserved)	7	RV3 (L/L)
		14	(Reserved)	6	FV3 (L/L)
		13	(Reserved)	5	MV3 (L/L)
		12	(Reserved)	4	FV2 (RHEED)
		11	(Reserved)	3	MV2 (RHEED)
		10	(Reserved)	2	FV1 (Main)
		9	(Reserved)	1	MV11 (Main bypass)
		8	(Reserved)	0	MV10 (Main)

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
50	Valve Stat2	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	(Reserved)
		12	(Reserved)	4	(Reserved)
		11	(Reserved)	3	(Reserved)
		10	(Reserved)	2	(Reserved)
		9	(Reserved)	1	(Reserved)
		8	(Reserved)	0	(Reserved)

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
51	Shut Stat	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	(Reserved)

		12	(Reserved)	4	(Reserved)
		11	(Reserved)	3	(Reserved)
		10	(Reserved)	2	(Reserved)
		9	(Reserved)	1	(Reserved)
		8	(Reserved)	0	Sample Shutter

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
52	Motor Stat	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	(Reserved)
		12	(Reserved)	4	(Reserved)
		11	(Reserved)	3	(Reserved)
		10	(Reserved)	2	(Reserved)
		9	(Reserved)	1	Target spin
		8	(Reserved)	0	Motor free

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
53	Other Stat	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	(Reserved)
		12	(Reserved)	4	(Reserved)
		11	(Reserved)	3	(Reserved)
		10	(Reserved)	2	(Reserved)
		9	(Reserved)	1	(Reserved)
		8	(Reserved)	0	Baking

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
54	A Utility 1	15	(Reserved)	7	(Reserved)

		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	(Reserved)
		12	(Reserved)	4	(Reserved)
		11	(Reserved)	3	(Reserved)
		10	(Reserved)	2	(Reserved)
		9	(Reserved)	1	Compressed air
		8	(Reserved)	0	Emergency shutdown

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
55	A Utility 2	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	(Reserved)
		12	(Reserved)	4	(Reserved)
		11	(Reserved)	3	(Reserved)
		10	(Reserved)	2	(Reserved)
		9	(Reserved)	1	(Reserved)
		8	(Reserved)	0	(Reserved)

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
56	A Pump 1	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	DP3 (L/L) Vac Error	5	(Reserved)
		12	DP2 (2nd RHEED) Vac Error	4	(Reserved)
		11	DP1 (Main) Vac Error	3	TMP21 (2nd RHEED) alarm
		10	DP3 (L/L) Thermal	2	TMP3 (L/L) alarm
		9	DP2 (2nd RHEED) Thermal	1	TMP20 (RHEED) alarm
		8	DP1 (Main) Thermal	0	TMP1 (Main) alarm

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
57	A Pump 2	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	(Reserved)
		12	(Reserved)	4	(Reserved)
		11	(Reserved)	3	(Reserved)
		10	(Reserved)	2	(Reserved)
		9	Break in baking temp. sensor	1	(Reserved)
		8	Tripped in baking heater line	0	(Reserved)

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
58	A EXT 1	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	(Reserved)
		12	Pyrometer error	4	(Reserved)
		11	Laser fiber head failure	3	(Reserved)
		10	Temperature sensor failure	2	(Reserved)
		9	Chiller (for heater) alarm	1	Door open in laser shield
		8	Heater power supply alarm	0	(Reserved)

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
59	A EXT 2	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	(Reserved)
		12	(Reserved)	4	(Reserved)
		11	(Reserved)	3	(Reserved)
		10	(Reserved)	2	(Reserved)
		9	(Reserved)	1	(Reserved)

		8	(Reserved)	0	(Reserved)
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No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
60	A Motor 1	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	Pulse failure in TG height (Z)
		12	Limit sensor in Mask2	4	Pulse failure in TG revolution
		11	Pulse failure in Mask2	3	(Reserved)
		10	Limit sensor in Mask1	2	(Reserved)
		9	Step out in L-mask encoder	1	(Reserved)
		8	Pulse failure in Mask1	0	Pulse failure in sample rotation

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
61	A Motor 2	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	Limit sensor in attenuator (R)
		12	(Reserved)	4	Pulse failure in attenuator (R)
		11	(Reserved)	3	Driver trouble in mirror (R)
		10	(Reserved)	2	Pulse failure in mirror (R)
		9	(Reserved)	1	Driver trouble in focus lens (R)
		8	(Reserved)	0	Pulse failure in focus lens (R)

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
62	W Pross	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	(Reserved)
		12	(Reserved)	4	(Reserved)
		11	(Reserved)	3	(Reserved)

		10	(Reserved)	2	Large drive-current deviation
		9	(Reserved)	1	Large temperature deviation
		8	(Reserved)	0	Miss-shot in excimer laser

No.	Data field	Bit	Status 0: OFF/1: ON	Bit	Status 0: OFF/1: ON
63	W EXT	15	(Reserved)	7	(Reserved)
		14	(Reserved)	6	(Reserved)
		13	(Reserved)	5	(Reserved)
		12	(Reserved)	4	(Reserved)
		11	(Reserved)	3	(Reserved)
		10	(Reserved)	2	(Reserved)
		9	(Reserved)	1	Mask1 confliction
		8	(Reserved)	0	T(Reserved)

5 Troubleshooting

5.1 Occurrence of accidents and their treatments in PLD system

PLD system shows alarm and beeps when the system has trouble in which the system cannot keep running during operation. Alarm and beep functions are set in the control rack (see Figure 18) and PC control software, a part of vacuum controller sub-window (see Figure 34). Finding trouble and its treatment can be done as follows in the software case.

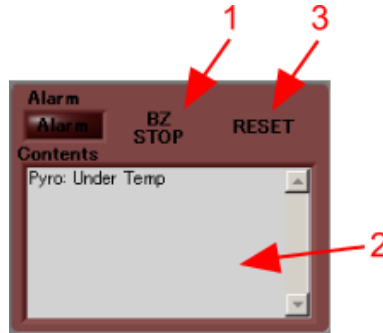


Figure 7:Error indicator part of vacuum controller sub-window in PC software.

(Troubleshooting procedure on PC control software)

- (1) Push "BZ STOP" button to stop beep.
- (2) Eliminate origins of the trouble according to the error message.
Error messages are shown by PLD's self diagnoses and interlock functions with prohibitive combination of operations.
- (3) Push reset button after elimination of trouble origins.

(Troubleshooting procedure using handy console)

- (4) Push "RESET" button to stop beep.
- (5) Eliminate origins of the trouble according to the error message.
Error messages are shown by PLD's self diagnoses and interlock functions with prohibitive combination of operations.
- (6) Push "RESET" button two times after elimination of trouble origins.

(Error clearance of motorized motion)

- (7) Error of the motorized motion parts are cleared by cyclic click of "Motor Stop" button with ON/OFF indication.
Error log is not cancelled in this case.

5.2 Error/warning messages

Table 3: Troubles of vacuum pumps/vacuum gauge in main chamber.

Error message	Level	Origins of troubles	Treatment for recovery
DP1 Main Thermal Tripped	Error / Alarm	Thermal relay switch for DP1 cuts electric power by compulsion due to overload or other accidents.	Get rid of the trouble origins in DP1.
DP1 CP (Circuit Protector) Tripped	Error / Alarm	Circuit protector switch for DP1 cuts electric power by compulsion due to circuit short or other accidents.	Get rid of the trouble origins in DP1.
DP1 (Main) Vacuum Error (TCG1 Setpoint not work)	Error / Alarm	Vacuum gauge controller cannot detect pressure-value set-point corresponding to turn-on condition in inter-lock function for TMP1 in 5 minutes.	Eliminate slow pumping in main chamber like vacuum leak, lowering DP1 performance or others.
TMP1 Main Alarm	Error / Alarm	TMP1 has troubles due to overload or other accidents.	Check operation status with error code in display at front panel of the TMP1 controller for recovery.
DP2 RHD Thermal Tripped	Error / Alarm	Thermal relay switch for DP2 cuts electric power by compulsion due to overload or other accidents.	Get rid of the trouble origins in DP2.
DP2 CP (Circuit Protector) Tripped	Error / Alarm	Circuit protector switch for DP2 cuts electric power by compulsion due to circuit short or other accidents.	Get rid of the trouble origins in DP2.
DP2 (Main) Vacuum Error (TCG2 Setpoint not work)	Error / Alarm	Vacuum gauge controller cannot detect pressure-value set-point corresponding to turn-on condition in inter-lock function for TMP1 in 5 minutes.	Eliminate slow pumping in main chamber like vacuum leak, lowering DP2 performance or others.
TMP20 RHD Alarm	Error / Alarm	TMP20 has troubles due to overload or other accidents.	Check operation status with error code in display at front panel of the TMP20 controller for recovery.
TMP21 RHD Alarm	Error / Alarm	TMP21 has troubles due to overload or other accidents.	Check operation status with error code in display at front panel of the TMP21 controller for recovery

Table 4: Troubles of vacuum pumps/vacuum gauge in load-lock chamber.

Error message	Level	Origins of troubles	Treatment for recovery
DP3 Main Thermal Tripped	Error / Alarm	Thermal relay switch for DP3 cuts electric power by compulsion due to overload or other accidents.	Get rid of the trouble origins in DP3.
DP3 CP (Circuit Protector) Tripped	Error / Alarm	Circuit protector switch for DP3 cuts electric power by compulsion due to circuit short or other accidents.	Get rid of the trouble origins in DP3.
DP3 (Main) Vacuum Error (TCG3 Setpoint not work)	Error / Alarm	Vacuum gauge controller cannot detect pressure-value set-point corresponding to turn-on condition in inter-lock function for TMP3 in 5 minutes.	Eliminate slow pumping in main chamber like vacuum leak, lowering DP3 performance or others.
TMP3 Main Alarm	Error / Alarm	TMP3 has troubles due to overload or other accidents.	Check operation status with error code in display at front panel of the TMP3 controller for recovery.

Table 5: Troubles in laser-diode substrate heater unit.

Error message	Level	Origins of troubles	Treatment for recovery
LD Power Supply Alarm	Error / Alarm	Laser-diode power supply has troubles in laser-diode module, cooling-water flow, or others.	Check alarm indicator at front panel of the laser-diode power-supply for recovery.
Chiller Unit Alarm	Error / Alarm	There is few cooling-water flow by chiller due to pump overload, shutdown, or others.	Check operation status of the chiller for recovery. Keep proper flow rate (5 liter/min) of cooling water after recovery.
LD Fiber Head Removed	Error / Alarm	Laser fiber is not properly connected at receptacle in the laser-optics unit.	Connect the signal cable.
Pyrometer Alarm	Error / Alarm	Pyrometer signal cable with pre-amplifier is not properly connected in the laser-optics unit.	Check operation status with error code in display at front panel of the TMP3 controller for recovery.
Sample Heating Temp. Check Error	Error / Alarm	Pyrometer cannot detect a sample temperature in set limit time due to misalignment of the laser beam path/pyrometer view path or low laser-diode performance.	Check the laser beam path or pyrometer view path alignment. Check the laser-diode performance.

Table 6: Troubles in motorized motion parts.

Error message	Level	Origins of troubles	Treatment for recovery
TG Select Motor Error	Error / Alarm	Stepping motor with its driver for target revolution cannot detect and control current/desired position.	Do homing operation after removing motor stack or other accidents.
Sample Rot. Motor Error	Error / Alarm	Stepping motor with its driver for sample rotation cannot detect and control current/desired position.	Reset (on PC) after removing motor stack or other accidents.
Mask Motor Error	Error / Alarm	Stepping motor with its driver for long-mask positioning cannot detect and control current/desired position.	Do homing operation after removing motor stack or resolving other problems.
Mask Motor Out Of Position	Error / Alarm	Sensor (rotary encoder) misses current position during long-mask operation.	Do reset and homing operation to clear position difference. Fix sensor mounting position.
Mask Motor Limit Sensor Worked	Error / Alarm	Long-mask reaches stroke limit.	Reverse the long-mask position.

Table 7: Troubles in chamber baking heater.

Error message	Level	Origins of troubles	Treatment for recovery
Baking ELB Tripped	Error / Alarm	ELB (Earth Leakage circuit Breaker) cuts electric power to tape heater for chamber baking.	Resolve tape heater problems like circuit short.
Baking TC Sensor Error	Error / Alarm	Thermocouple (TC) temperature sensor cannot measure a chamber heating temperature.	Resolve TC (compensative) sensor cable problems like break, Stop overheating.

Table 8: Forced stop in PLD system.

Error message	Level	Origins of troubles	Treatment for recovery
Air Low Pressure	Error / Alarm	Pressure of the compressed air is not enough for pneumatic valve operations.	Resolve compressed air problems including air source (compressor), piping, and regulator.
Emergency Stop	Error / Alarm	Operator pushes emergency off (EMO) button.	Release the EMO-button lock after clearing troubles in the PLD system.

6 System status in emergency circumstances

6.1 Status at cut off the electricity

Status of the PLD system at electrical shutdown is as follows.

(1) Pumps

All turbo molecular pumps (TMPs) stop due to no power supply. Rotation speeds of the TMPs decrease by natural braking. In the case of TMP1 the pose of the rotor by magnetic levitation is kept by self power supply and then a bearing touches down to a guide when the self power supply stops.

All rotary/scroll pumps stop due to no power supply.

(2) Valves

Some valves by pneumatic operation keeps current status (usually open) until the compressed air goes down to critical pressure at around 4-5 MPa. Then the status gets uncertain because of lack of the compressed air. Other valves keeps current status by manual operation. All valves do not close automatically at the electrical shutdown.

(3) Control rack part

All components and units turn off because of the electrical shutdown.

(4) Computer

The computer turns off because of the electrical shutdown. In the system there is no UPS unit.

(5) Ablation laser

The ablation laser turns off because of the electrical shutdown.

6.2 Status at pushing emergency button

Pushing the emergency button causes the electric shutdown in the control tower. So the status of the components or the units are similar to the electrical shutdown described above. Only the ablation laser cannot be controlled by the emergency button.

6.3 Recovery

(1) Short interruption time

If it is obvious that the PLD chamber is kept UHV for short interruption time, only turning on all components and units is enough to become normal operation / status.

(2) Long interruption time

If it is not obvious that the PLD chamber is kept UHV for long interruption time, do the operations described in section 4-4. Do evacuation process again.

Maintenance manual of PAC-LMBE system

Rev 1 March. 15, 2024

Pascal Co., Ltd.

Headoffice

1-16-4 Showa-cho, Abeno-ku, Osaka 545-0011, Japan

TEL : +81-6-6626-1321 FAX : +81-6-6626-1323

Contact-address pascal@pascal-co-ltd.co.jp

